INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

GASKET AND PACKING MATERIAL, RUBBER, FOR USE WITH POLAR FLUIDS, STEAM, AND AIR AT MODERATELY HIGH TEMPERATURES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rubber gaskets and packing material, seals, strips, molded forms, and sheet material for use with hot air, hot water, steam, and various polar fluids, such as aqueous monoethanolamine (MEA), at moderately high temperatures (see 6.1).

1.2 Classification. The rubber gasket and packing material are of the following grades, as specified (see 6.2).

a. Grade 1 – Nominal 65 durometer hardness.

b. Grade 2 – Nominal 80 durometer hardness.

c. Grade 3 – Nominal 90 durometer hardness.

1.3 Part or identifying number (PIN). PINs to be used for rubber gasket and packing material acquired to this specification are created as follows:

M	**22050**	**-**	**X**	**-**	**XXX**
Prefix for Military Specification	Specification Number		Grade 1, 2, or 3 (see 1.2)		SAE AS568 dash number for size designation

Example: M22050-1-001

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5330

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-M-23573 - Monoethanolamine Chelating Agent Solution

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 261 - Identification and Listing of Hazardous Waste

(Copies of this document are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D512 - Standard Test Methods for Chloride Ion in Water

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

ASTM D3767 - Standard Practice for Rubber – Measurement of Dimensions

ASTM E662 - Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

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ASTM E800 - Standard Guide for Measurement of Gases Present or Generated During Fires

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3601-3 - Fluid Power Systems – O-rings – Part 3: Quality Acceptance Criteria

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AS568 - Aerospace Size Standard for O-rings

SAE AS5752 - Aerospace - Visual Inspection Standard for Elastomeric Sealing Elements Other than O-Rings

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be made from an ethylene-propylene elastomeric compound that shall meet the applicable requirements specified herein. Suggested formulations are given in 6.4 for both grade 1 and grade 2 material. The material shall contain no low-melting-point metals such as mercury, lead, or antimony in its formulation. The material shall be furnished in the form of sheets, strips, gaskets, packings, seals, or molded items of the sizes specified (see 6.2).

3.2 O-rings. Unless otherwise specified (see 6.2), the required O-ring shall be identified by its PIN (see 1.3) and shall have the dimensions and tolerances in accordance with the requirements of the SAE AS568 dash number.

3.3 Physical requirements of the rubber. The rubber compound used in the gaskets and packing material shall conform to the applicable requirements in table I when tested as specified herein.

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TABLE I. Physical requirements of rubber compounds.

		Grade 1	Grade 2	Grade 3
Initial properties	Tensile strength, pounds per square inch, minimum (min)	2,400	2,400	2,200
	Ultimate elongation, percent, min	250	150	80
	Hardness, Shore A durometer, points within 1 second	65±5	80±5	90±5
	Leachable chlorides, parts per million, maximum (max)	200	200	200
Properties after oven aging 166 hours at 280 °F	Tensile strength, percent of initial, min	70	70	70
	Ultimate elongation, percent of initial, min	70	70	70
	Hardness, change in Shore A durometer, points max	15	15	15
	Compression set, percent, max	60	60	35
Properties after immersion in distilled water for 166 hours at 330 °F	Tensile strength, percent of initial, min	80	80	80
	Ultimate elongation, percent of initial, min	80	80	80
	Volume change, percent (no shrinkage allowed)	0 to 10	0 to 10	0 to 10
Properties after immersion in 25 percent aqueous solution of MEA for 166 hours at 330 °F	Tensile strength, percent of initial, min	80	80	--
	Ultimate elongation, percent of initial, min	80	80	--
	Volume change, percent (no shrinkage allowed)	0 to 10	0 to 10	--
Properties after 166 hours in steam at 330 °F	Tensile strength, percent of initial, min	80	80	80
	Ultimate elongation, percent of initial, min	80	80	80
	Volume change, percent, max	5	5	5
Properties after 70 hours in steam at 390 °F	Tensile strength, percent of initial, min	--	--	90
	Ultimate elongation, percent of initial, min	--	--	85
	Hardness, change in Shore A durometer, points max	--	--	3
	Volume change, percent (no shrinkage allowed)	--	--	0 to 8
	Compression set, percent, max	--	--	60
Properties after 166 hours in steam at 550 °F	Tensile strength, percent of initial, min	--	--	75
	Ultimate elongation, percent of initial, min	--	--	15
	Hardness, change in Shore A durometer, points max	--	--	10
	Volume change, percent (no shrinkage allowed)	--	--	0 to 5
	Compression set, percent, max	--	--	90

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3.4 Identification. Material acquired under this specification shall be identified with white markings. A permanent marking material that will have no deleterious effect on the rubber shall be used, or, in the case of individual one-item packaging, the marking may be put on the package.

3.5 Workmanship. The material shall be examined for defects as specified herein.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Toxicity and prohibited materials.

3.7.1 Toxicity. When evaluated in accordance with 4.4.1, the gasket and packing material shall pose no serious risk to the health of personnel or the environment when used for its intended purpose (see 4.4.1 and 6.6).

3.7.2 Prohibited materials. The gasket and packing material shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.8 Disposal. The gasket and packing material shall not contain any hazardous material or exhibit any hazardous characteristic that would cause the material to be designated as hazardous waste as defined under 40 CFR 261. Alternatively, see 6.2 to support the disposal of hazardous waste.

3.9 Off-gassing. Unless otherwise specified (see 6.2), the gasket and packing material shall be evaluated for off-gassing in accordance with the requirements of 4.5. Based on the circumstances of use and the chemical nature of the gasket and packing material, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the gasket and packing material shall be assigned to either the "Permitted" or "Limited" category (see 4.5 and 6.7).

3.10 Fire gas toxicity. The gasket and packing material shall be evaluated for fire gas toxicity in accordance with ASTM E662 and ASTM E800.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in 4.2.

4.1.1 Lot. For the purpose of conformance inspection and testing, a lot is defined as all the products of the same grade and materials made in a single facility under the same production processes that are being offered for delivery at one time. A lot shall not exceed 1,000 pounds.

4.1.2 Sampling for conformance inspection. A random sample of gasket and packing material shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.2 Examinations and tests.

4.2.1 Visual and dimensional examination. Each of the samples taken in accordance with 4.1.2 shall be subjected to surface examinations for identification, workmanship, dimensions, and tolerances. Non-O-ring elastomeric seals shall meet the acceptance criteria in accordance with SAE AS5752 and O-rings shall meet the acceptance criteria in accordance with ISO 3601-3. The dimensions shall be determined in accordance with the applicable method specified in ASTM D3767.

4.2.2 Conformance tests. Conformance tests shall be conducted on samples selected in accordance with 4.1.2. Conformance tests shall consist of the examination and tests specified in 4.3 (see 6.3).

4.2.3 Action in case of nonconformance. If any of the samples in the conformance tests are found not to be in conformance with the requirements of the specification, the lot that they represent shall be rejected.

4.3 Test methods.

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4.3.1 Testing conditions. Unless otherwise specified herein, the testing shall be conducted at a temperature of 80±9 °F.

4.3.2 Tensile properties. The tensile strength and ultimate elongation shall be determined in accordance with ASTM D412. Die C specimens that are 0.08±0.01 inch thick shall be used for determinations of tensile properties.

4.3.3 Hardness. Hardness shall be determined with a Shore A durometer on specimens that are 0.5±0.01 inch thick and at least 1.129 inches in diameter in accordance with ASTM D2240. The hardness readings shall be taken within 1 second after firm contact is made between the rubber and the presser foot of the durometer.

4.3.4 Oven aging. The test specimens shall be oven aged in accordance with the ASTM D573 method specified for each test, except that the aging period shall be 166±1 hours at 280±2 °F. After the aging period, the tensile hardness specimens shall be set aside for 16 to 96 hours at 80±9 °F before determining the required property.

4.3.4.1 Tensile properties after oven aging. The tensile properties after oven aging shall be determined by the procedure specified in 4.3.2.

4.3.4.2 Hardness after oven aging. The hardness after oven aging shall be determined by the procedure specified in 4.3.3.

4.3.4.3 Compression set after oven aging. Compression set shall be determined in accordance with ASTM D395, except that three 0.5±0.01-inch-thick specimens shall be tested and the specimens shall be held at a 25 percent deflection during oven aging.

4.3.5 Immersion in hot distilled water. Test specimens for the individual tests shall be immersed in distilled water in a covered glass jar. The jar shall be put in a container capable of withstanding an internal pressure of 300 pounds per square inch (lb/in^2). The container shall be sealed and placed in a steam autoclave and heated for 166±1 hours at 330±2 °F.

4.3.5.1 Tensile properties after immersion in hot distilled water. After immersion in hot distilled water as specified in 4.3.5, the tensile properties shall be determined in accordance with ASTM D471 and as specified in 4.3.2. Tensile strength shall be based on the swollen cross-sectional area.

4.3.5.2 Volume change after immersion in hot distilled water. Volume change after immersion in hot distilled water shall be determined in accordance with ASTM D471.

4.3.6 Immersion in 25 percent aqueous solution of MEA. Test specimens for the individual tests shall be immersed in a 25 percent by volume aqueous solution of MEA in a covered glass jar. The jar shall be put in a container capable of withstanding an internal pressure of 300 lb/in^2. The container shall be sealed and placed in a steam autoclave and heated for 166±1 hours at 330±2 °F. The MEA solution shall be in accordance with MIL-M-23573. These tests shall not be applied to grade 3 material.

4.3.6.1 Tensile properties after immersion in MEA solution. After immersion in a 25 percent aqueous solution of MEA (see 4.3.6), the tensile properties shall be determined as specified in 4.3.5.1.

4.3.6.2 Volume change after immersion in MEA solution. After immersion in a 25 percent aqueous solution of MEA (see 4.3.6), the volume change shall be determined in accordance with ASTM D471.

4.3.7 Aging in steam.

4.3.7.1 Aging in steam at 330 °F. Test specimens for the individual tests shall be exposed to steam in an autoclave for 166±1 hours at 330±2 °F. After the aging period, the tensile specimens shall be set aside at 80±9 °F for 16 to 96 hours before determining the required property.

4.3.7.1.1 Tensile properties after aging in steam at 330 °F. After aging in steam at 330 °F (see 4.3.7.1), the tensile properties shall be determined as specified in 4.3.2.

4.3.7.1.2 Volume change after aging in steam at 330 °F. After aging in steam at 330 °F (see 4.3.7.1), the volume change shall be determined in accordance with ASTM D471.

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4.3.7.2 Aging in steam at 390 °F. Test specimens for the individual tests shall be exposed to steam in an autoclave for 70±1 hours at 390±2 °F. After the aging period, the tensile specimens shall be set aside at 80±9 °F for 16 to 96 hours before determining the required property.

4.3.7.2.1 Tensile properties after aging in steam at 390 °F. After aging in steam at 390 °F (see 4.3.7.2), the tensile properties shall be determined as specified in 4.3.2.

4.3.7.2.2 Hardness after aging in steam at 390 °F. After aging in steam at 390 °F (see 4.3.7.2), the hardness shall be determined as specified in 4.3.3.

4.3.7.2.3 Volume change after aging in steam at 390 °F. After aging in steam at 390 °F (see 4.3.7.2), the volume change shall be determined in accordance with ASTM D471.

4.3.7.2.4 Compression set after aging in steam at 390 °F. After aging in steam at 390 °F (see 4.3.7.2), the compression set shall be determined in accordance with ASTM D395, except that three 0.5±0.01-inch-thick specimens shall be tested and the specimens shall be held at a 25 percent deflection during steam aging.

4.3.7.3 Aging in steam at 550 °F. Test specimens for the individual tests shall be exposed to steam in an autoclave for 166±1 hours at 550±2 °F. After the aging period, the tensile specimens shall be set aside at 80±9 °F for 16 to 96 hours before determining the required property.

4.3.7.3.1 Tensile properties after aging in steam at 550 °F. After aging in steam at 550 °F (see 4.3.7.3), the tensile properties shall be determined as specified in 4.3.2.

4.3.7.3.2 Hardness after aging in steam at 550 °F. After aging in steam at 550 °F (see 4.3.7.3), the hardness shall be determined as specified in 4.3.3.

4.3.7.3.3 Volume change after aging in steam at 550 °F. After aging in steam at 550 °F (see 4.3.7.3), the volume change shall be determined in accordance with ASTM D471.

4.3.7.3.4 Compression set after aging in steam at 550 °F. After aging in steam at 550 °F (see 4.3.7.3), the compression set shall be determined in accordance with ASTM D395, except that three 0.5±0.01-inch-thick specimens shall be tested and the specimens shall be held at a 25 percent deflection during steam aging.

4.3.8 Water-leachable chlorides. The following test methods shall be used:

a. Two test specimens, each of approximately, but not less than, 14 grams, shall be taken. Each test specimen shall be taken from a different sample. The test specimens for rubber-bonded paper shall be taken from the samples of the first and the last roll of the lot. The test specimen of packing material shall be representative of the sample cross-section.

b. The test specimens shall be unraveled and cut into pieces no larger than ¼ by ½ inch.

c. Each test specimen shall be weighed, transferred to an Erlenmeyer flask equipped with a reflux condenser, covered with 200 to 300 milliliters of distilled or demineralized water, and simmered in the range of 200 to 212 °F for a minimum of 6 hours. The leach water shall then be separated by filtration and the filter shall be rinsed. A blank determination shall be run using similarly cleaned equipment and distilled or demineralized water from the same source.

d. The extracted chloride ion shall be determined by the mercuric nitrate titration method in accordance with ASTM D512 referee method A, or equal, and shall be corrected by the results of the blank determination.

e. The results of each test shall be calculated as net parts per million of extractable chloride ion by weight of the test specimen.

4.4 Toxicity and prohibited materials.

4.4.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.7.1 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the gasket and packing material using data provided by the manufacturer/distributor to the NMCPHC (see 3.7.1 and 6.6).

4.4.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.7.2 and 6.2).

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4.5 Off-gassing. The gasket and packing material shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.9 and 6.7). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.9 and 6.11). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.9).

4.6 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.8).

4.7 Fire gas toxicity. During the conduct of the smoke density test specified in 3.10, the gasket and packing material shall be evaluated for fire gas toxicity in accordance with ASTM E800. The average value of the maximum value of the gas concentration measured at each condition (flaming and non-flaming) shall be recorded for the following gases, as specified (see 6.2): carbon monoxide (CO), hydrogen chloride (HCl), hydrogen fluoride (HF), nitrogen oxides (NOx), hydrogen bromide (HBr), hydrogen cyanide (HCN), and sulfur dioxide (SO_2). When making toxicity measurements, the sampling of gases shall be made during the testing of the second and the third specimen at each test condition, from the geometrical center of the chamber at the time when the maximum specific optical density of smoke is reached. The number of samples shall be four: two each in the flaming and non-flaming condition. The concentration of each toxic gas shall be determined as parts per million in the chamber volume.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material furnished under this specification is intended for use in air at temperatures up to 230 °F and in water, steam, and certain polar fluids at temperatures up to 280 °F. This material is especially intended for use in CO_2 scrubbers, in high-discharge systems, and in flexible joints. The suitability of this material for use in fluids other than those designated in this specification is determined beforehand in appropriate immersion tests. The grades and applications of the material are as follows:

a. Grade 1 material is intended for use in all applications in 1.1, at temperatures up to and including 230 °F.

b. Grades 2 and 3 materials are intended for use in all applications listed in 1.1, at temperatures up to and including 280 °F, except that grade 3 is not intended for use in various polar fluids such as aqueous MEA.

This product must survive in a maritime environment in excess of commercial standards and operate at higher levels of reliability to meet the Navy's mission.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade of material required (see 1.2).

c. Sizes specified for material (see 3.1).

d. O-ring PIN, dimensions, and tolerances, if other than specified (see 3.2).

e. Requirement for prohibited materials verification (see 3.7.2 and 4.4.2).

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f. If the manufacturer is unable to provide a product that complies with 3.8 (i.e., if the gasket and packing material contain a hazardous material), then the manufacturer should provide information detailing proper disposal of the gasket and packing material in compliance with 40 CFR 261.

g. Off-gassing evaluation methods, if other than specified (see 3.9 and 6.7).

h. AQL of 1.0 (see 4.1.2).

i. Requirements for an HHA (see 4.4.1 and 6.6).

j. Additional fire testing provisions (see 4.6 and 6.8).

k. Requirements for fire gas toxicity (see 3.10, 4.7, and 6.9).

l. Packaging requirements (see 5.1 and 6.10).

m. Safety Data Sheet (SDS), when required (see 6.5).

6.3 Limited testing option. Conformance testing can be reduced, provided the following criteria are met: If the product has been previously tested and the test results verified in writing by the Defense Contract Management Agency (DCMA), the following tests from 4.3 can be omitted from the conformance inspection: 4.3.5, 4.3.6, 4.3.7, and 4.3.8.

6.4 Suggested formulation. The following formulations are offered only as suggestions. The Government does not guarantee that materials made from these formulations will conform to the requirements of this specification. Use of these formulations does not relieve the supplier from having his product tested for conformity with the specification. Vulcanizates prepared from the formulation shown in table II have been found to meet the requirements of this specification:

TABLE II. Suggested formulations.

Compounding ingredients	Grade 1 parts by weight	Grade 2 parts by weight
Ethylene propylene diene monomer polymer	100	100
N358 Carbon black	50	80
Zinc oxide	5	5
Calcium stearate	1	1
Polymerized 1,2-dihydro-2,2,4-trimethylquinoline	2	2
Sulfur	0.3	0.3
Dicumyl peroxide	10	10
NOTE: Cure – 30 minutes at 320 °F (160 °C).		

6.5 Safety data sheets. Contracting officers will identify those activities requiring copies of completed Safety Data Sheets prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract (see 6.2).

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

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6.7 Off-gassing. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the atmosphere and results in health hazards to personnel or deleterious effects on machinery. These controls are accomplished through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010, chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.8 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. The Technical Authority will provide a letter upon request that indicates the approved laboratories which are accredited to ISO/IEC 17025 and that details how to obtain and maintain accreditation (see 6.2).

6.9 Fire gas toxicity evaluation. NAVSEA recommends inclusion of the fire gas toxicity evaluation (see 3.10 and 4.7). Measurement of the concentration of each toxic gas should be as prescribed in 4.7 and provided in the first article test report (see 6.2). The Navy will use the data provided for evaluation purposes.

6.10 Preservation and packaging materials. Gasket preservation and packaging materials should contain no low-melting-point metals such as mercury, lead, or antimony (see 6.2).

6.11 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. Contact NAVSEA 05S for assistance with determining the appropriate Technical Authority for U.S. Navy ships.

6.12 Subject term (key word) listing.

Aqueous menothanolamine

Ethylene-propylene elastomeric

Molded

O-rings

Seals

Sheets

Strips

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 5330-2018-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:59Z
Check the source to verify that this is the current version before use.